

Responding to a Zoonotic Disease Outbreak - Epidemiological interpretation and implications for decision-making in One Health

Afternoon hands-on tutorial

2026-07-07

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1 Overview

We will explore surveillance data from Integrated Bite Case Management (IBCM) in MIMAROPA region. IBCM is a One Health approach to rabies surveillance that links post-exposure vaccination of bite victims, investigation of biting animals, community sensitisation and programmatic coordination, to improve prevention, detection and response. IBCM was introduced to Romblon and Oriental Mindoro provinces in 2020, although COVID-19 restrictions prevented full implementation. In 2022 IBCM expanded to Marinduque province and in 2024 to Occidental Mindoro.

Using these data you will examine where outbreaks occurred; how these outbreaks spread; and why. In groups you will look more deeply into one province and consider the implications for control and prevention.

2 Learning objectives

In this session you will:

- 1) gain understanding of how to **critically interpret epidemiological data** considering a **One Health perspective** and the **quality of evidence**.
- 2) In so doing you should: **identify gaps in surveillance and control**; and **make recommendations** for future action.

Code is folded but available via the **Show code** button.

i Files you need (in your workshop folder)

- `dat_all_anon.csv` — human and animal rabies cases for MIMAROPA
- `mdv_summary.csv` — dog vaccination data (timing of vaccination campaigns in the region by province)
- `rolling_vac_cov_yr.csv` — hupreprocessed estimates of vaccination coverage by municipality
- `PHL_municipality.shp` (and its `.dbf`, `.shx`, `.prj` files) — municipality polygons for MIMAROPA

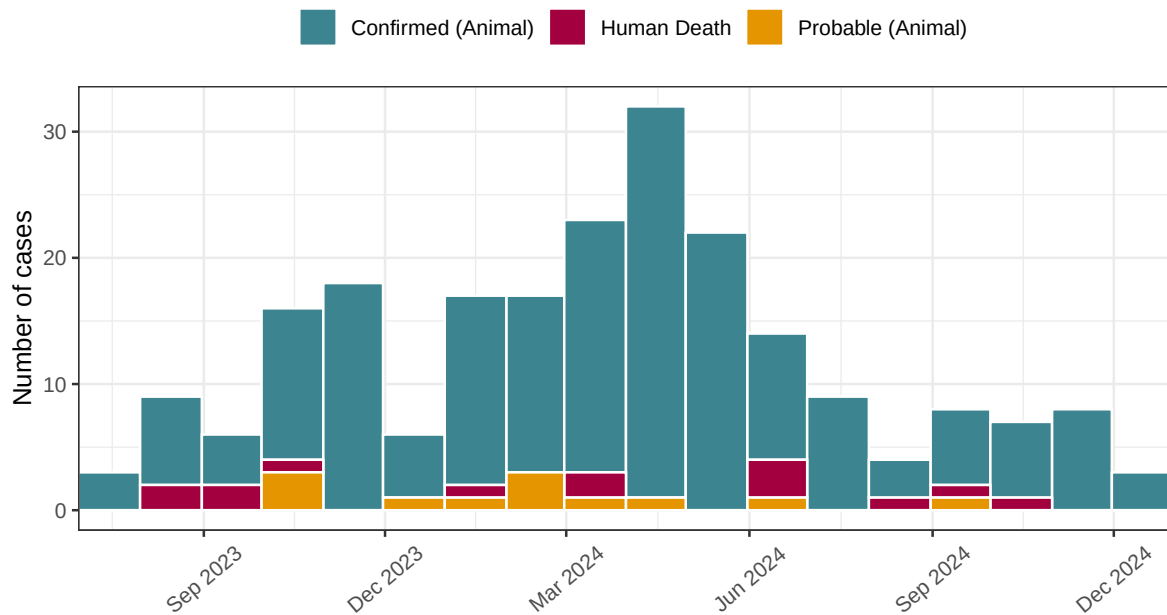
i import and process the data for plotting

```
# A tibble: 6 x 9
  barangay      municipality province case_status date case_status_fct case_type
  <chr>         <chr>         <chr>   <chr>    <chr> <chr>         <chr>
1 Apitong      Boac           Marindu~ Confirmed 24/0~ Confirmed (Ani~ Animal
2 Balogo       Boac           Marindu~ Confirmed 10/1~ Confirmed (Ani~ Animal
3 Agumaymayan Boac           Marindu~ Confirmed 06/1~ Confirmed (Ani~ Animal
4 Laylay       Boac           Marindu~ Confirmed 23/1~ Confirmed (Ani~ Animal
5 Bangbangan Boac           Marindu~ Probable 13/1~ Probable (Anim~ Animal
6 Bangbangan Boac           Marindu~ Probable 16/1~ Probable (Anim~ Animal
# i 2 more variables: year <dbl>, island <chr>
```

3 Describing the outbreak:

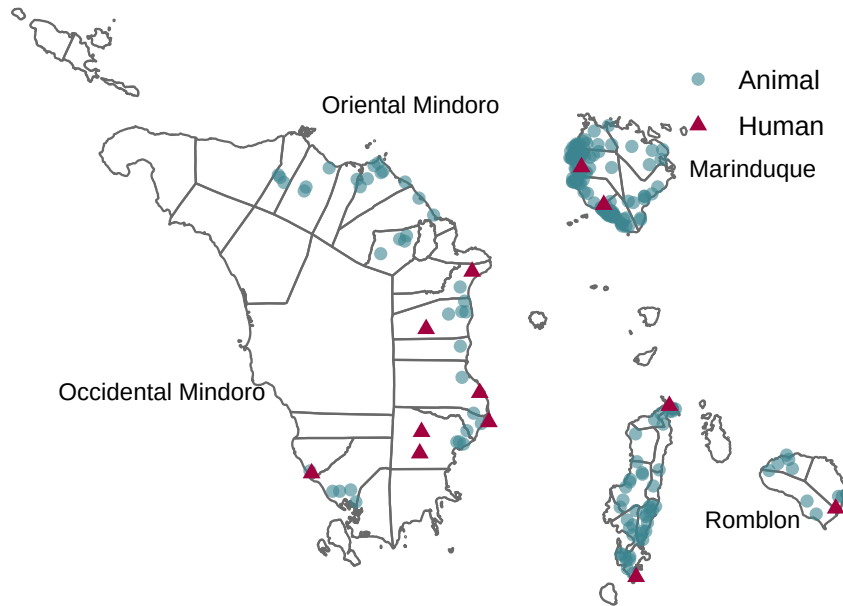
3.1 Epidemic curve

Monthly reported animal rabies cases (confirmed and probable) and human rabies deaths in Oriental Mindoro, Romblon, Marinduque and Occidental Mindoro are presented in a time series.



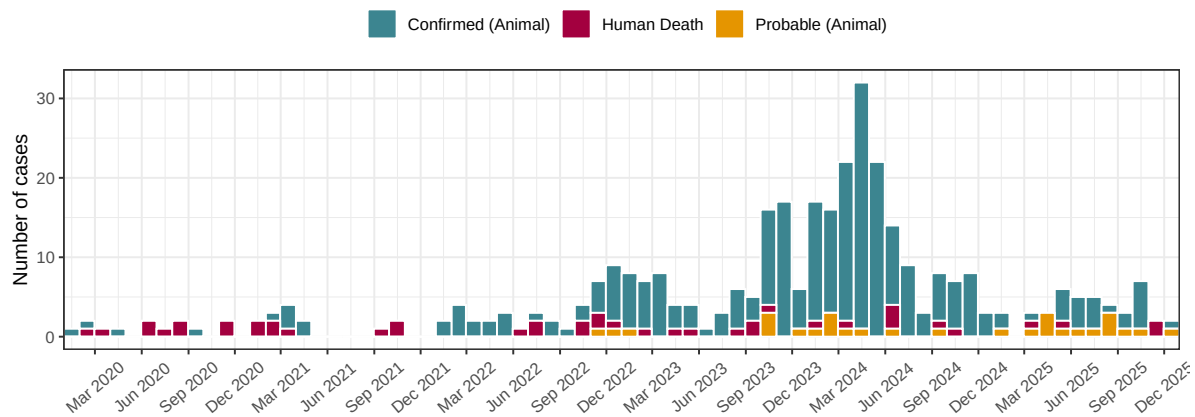
3.2 Locations of cases

The spatial distribution of all these cases are mapped. Human deaths are shown as triangles, whilst animal cases are shown as circles.

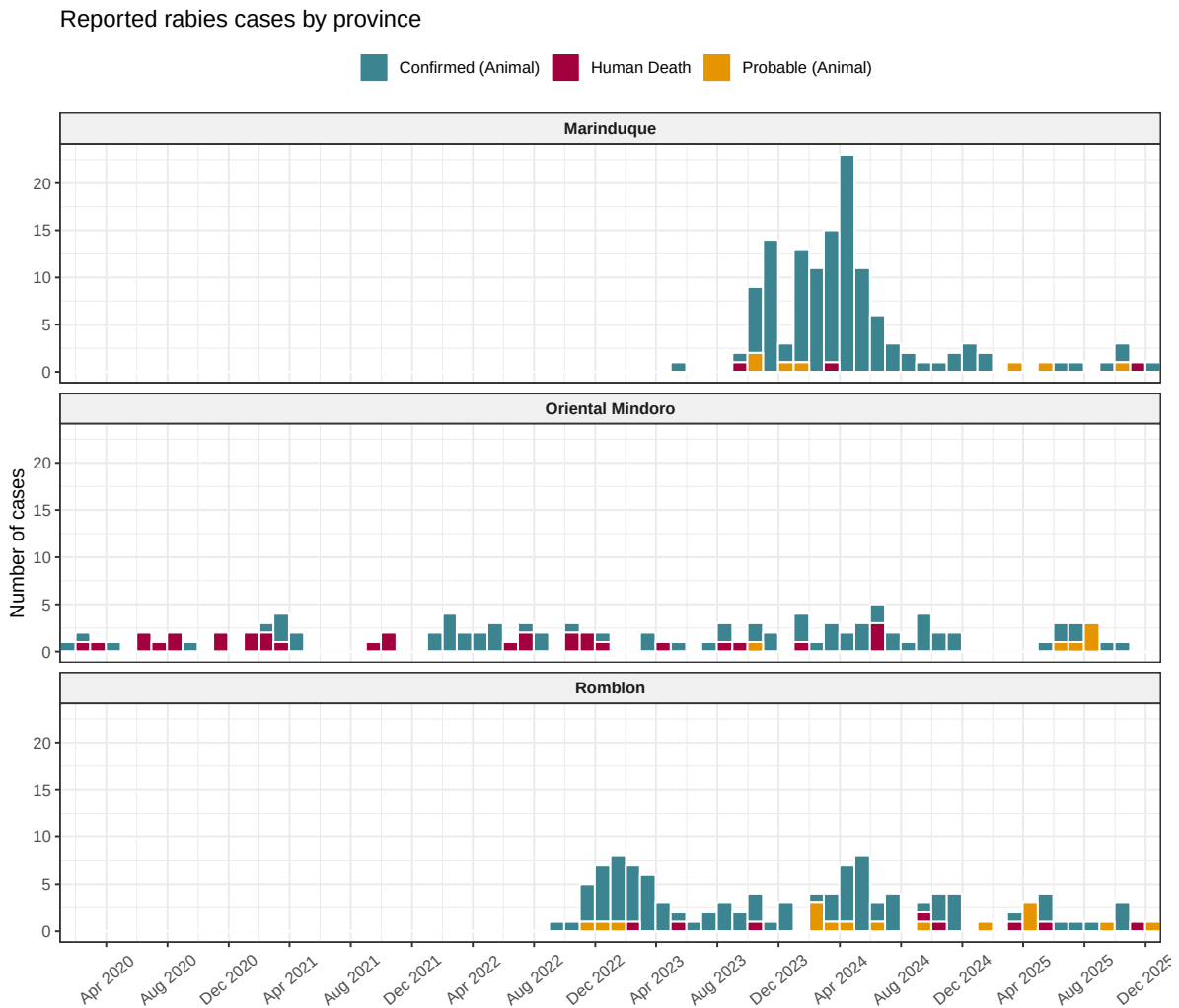


3.3 Take a closer look at the extended epidemic curve(s) for the region

Monthly reported animal cases (confirmed and probable) and human deaths in Oriental Mindoro, Romblon, Marinduque and Occidental Mindoro including all data from 2019 - 2026

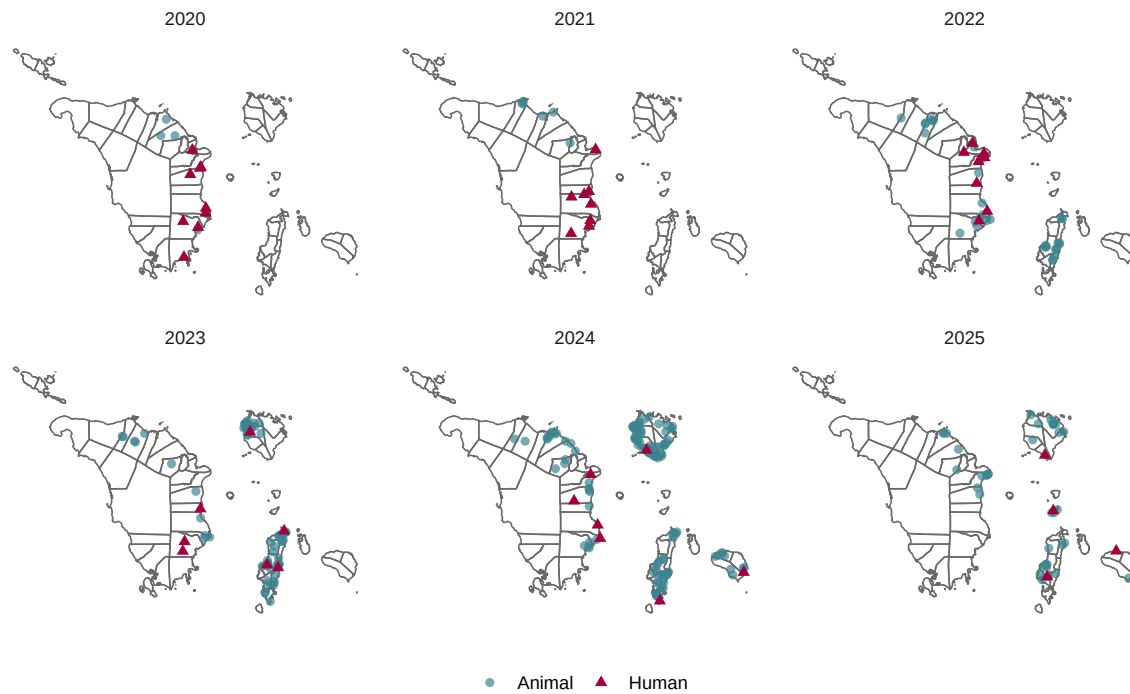


3.4 Plot the epidemic curve by province



3.5 Look at the locations of cases across the years

Spatial distribution of all cases from 2019 to 2025. Human deaths are shown as triangles, whilst animal cases are shown as circles.



4 How did the epidemiological situation evolve in each province?

Looking at the epidemic curve by province reveals whether provinces were affected simultaneously or sequentially — this is important for understanding routes of disease spread and the timing of introduction events

💡 Using your R skills work through the data in your Provincial groups!

Add the data on mass dog vaccination campaigns to help with your interpretation

4.1 Oriental Mindoro

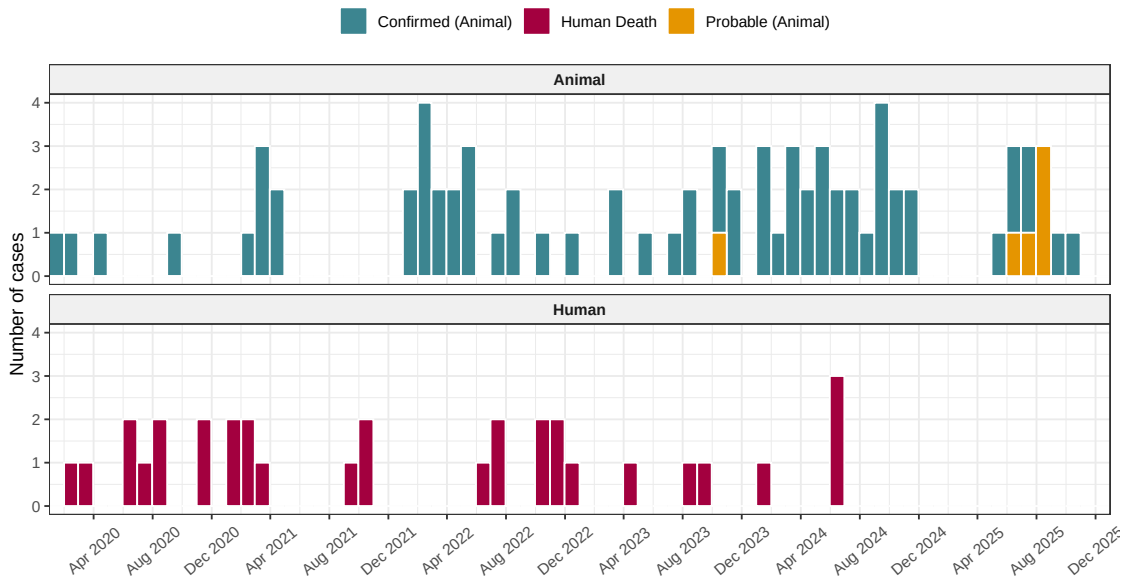
Key questions and things to explore:

- What is happening in the early outbreak period?
- How do animal cases compare to human cases in Oriental Mindoro?
- Roughly 1 person is bitten every 3 rabid dogs. Of people bitten, 90% get PEP. Of the 10% that don't get PEP, 20% will develop rabies and die. How does this change your interpretation of the early outbreak period?
- How does the most recent human death in April 2026 (look at the 2026 map!) change your interpretation of what is happening in Oriental Mindoro?

4.1.1 How do human cases compare to animal cases in Oriental Mindoro?

You will need to filter the outbreak data for Oriental Mindoro province and use the facet command to split these data according to human and animal cases.

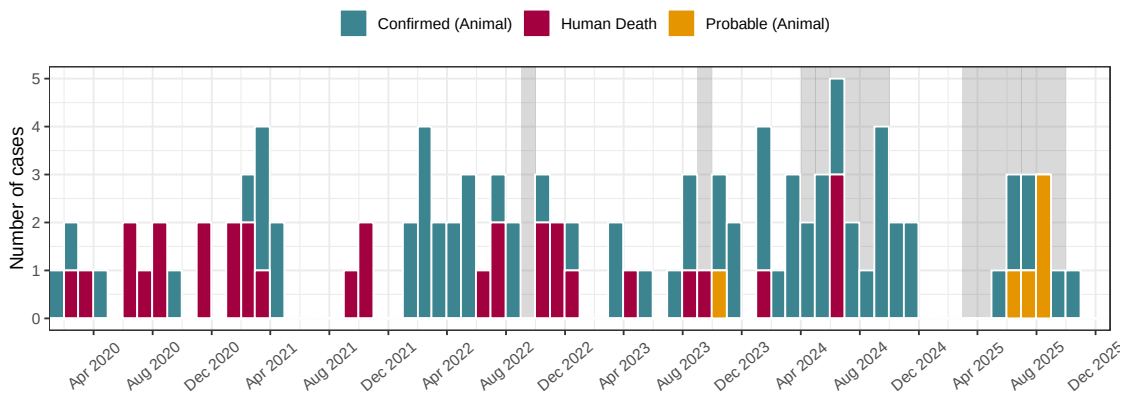
💡 Show solution



4.1.2 What impact do you think the mass dog vaccination campaigns had on rabies transmission?

Overlay the mass dog vaccination data to the timeseries plot. The mass dog vaccination data will need filtering too!

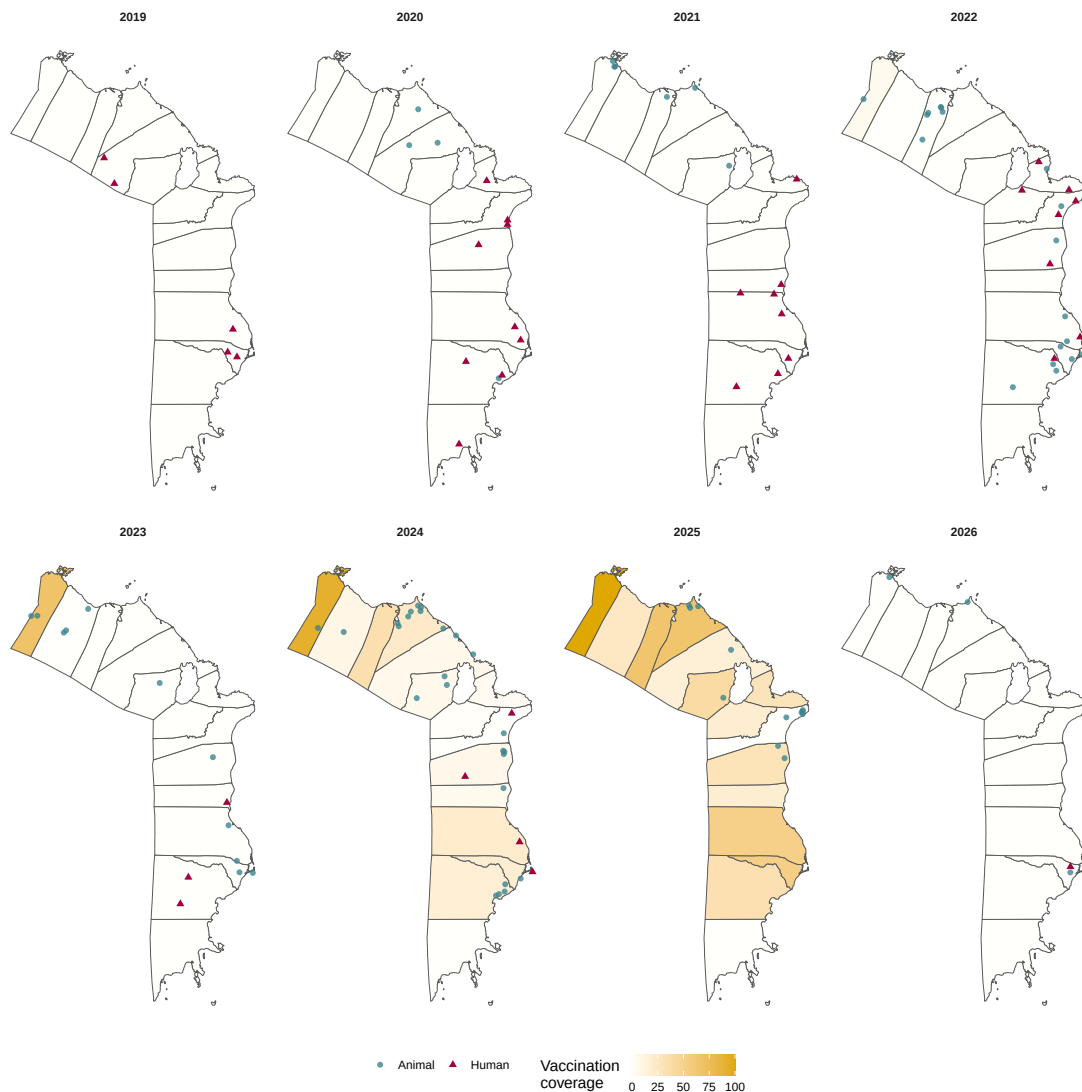
💡 Show solution



4.1.3 Does the distribution of cases over time and across municipalities suggest any gaps in surveillance (from the health or animal health perspective)?

Try joining the Oriental mindoro data with municipality vaccination data - it is a little bit tricky!

💡 Show solution



Note: some vaccination has occurred in 2026, however this data has not yet been released

4.2 Romblon

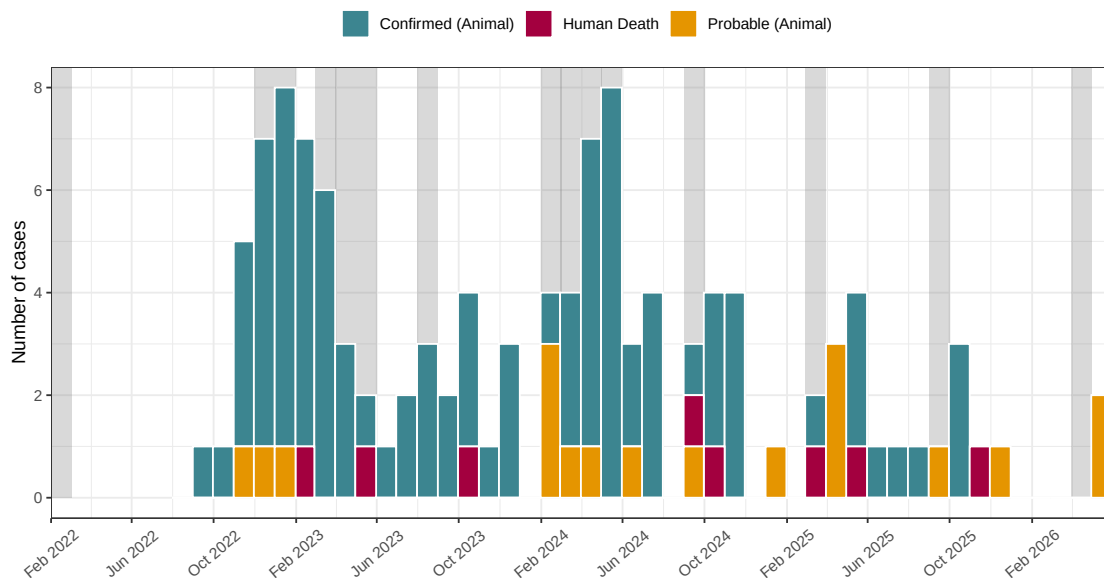
Key questions and things to consider:

- Romblon province is made up of different islands. How did rabies spread across the islands over time?
- If you were facing the situation in Tablas island in December 2022 (look at the 2022 map), what actions would you recommend to the local authorities?
- When did the first death occur? What can you say about the responsiveness of dog vaccination efforts?
- What if you were facing the situation in Sibuyan island in December 2023, would you be concerned? what communications would you want between municipalities? How might your thinking change by August 2024? Do you have any concerns if you are from Corcuera island? What do these complete plots tell you about the current surveillance efforts in Romblon?

4.2.1 How does the outbreak look in Romblon if we break this down by island?

You will need to filter the outbreak data to Romblon Province. Overlay the mass dog vaccination data to the timeseries plot. The mass dog vaccination data will need filtering too!

 Show solution

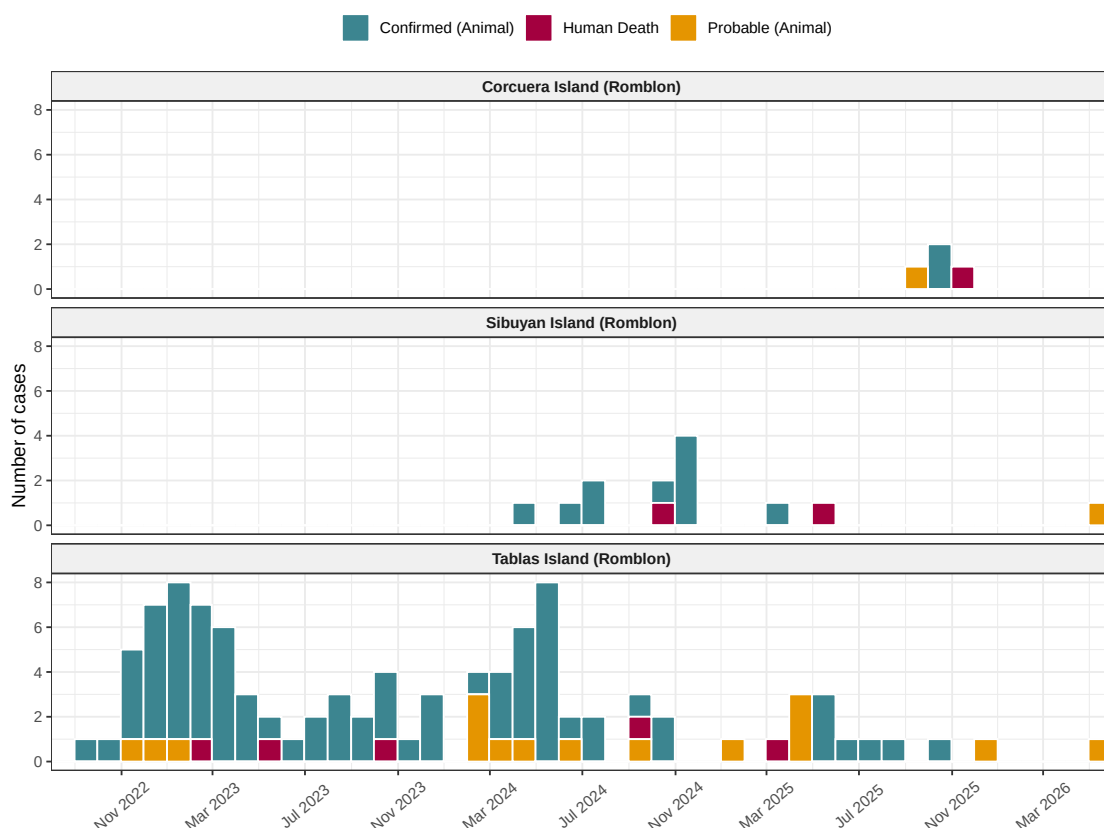


Dates of mass dog vaccination campaigns are shown in grey (MDV campaign defined as >700 dogs vaccinated in a municipality in a month)

4.2.2 Can you show the epidemic curves for each of Romblon's islands?

Use the facet command to split up the data by island

💡 Show solution

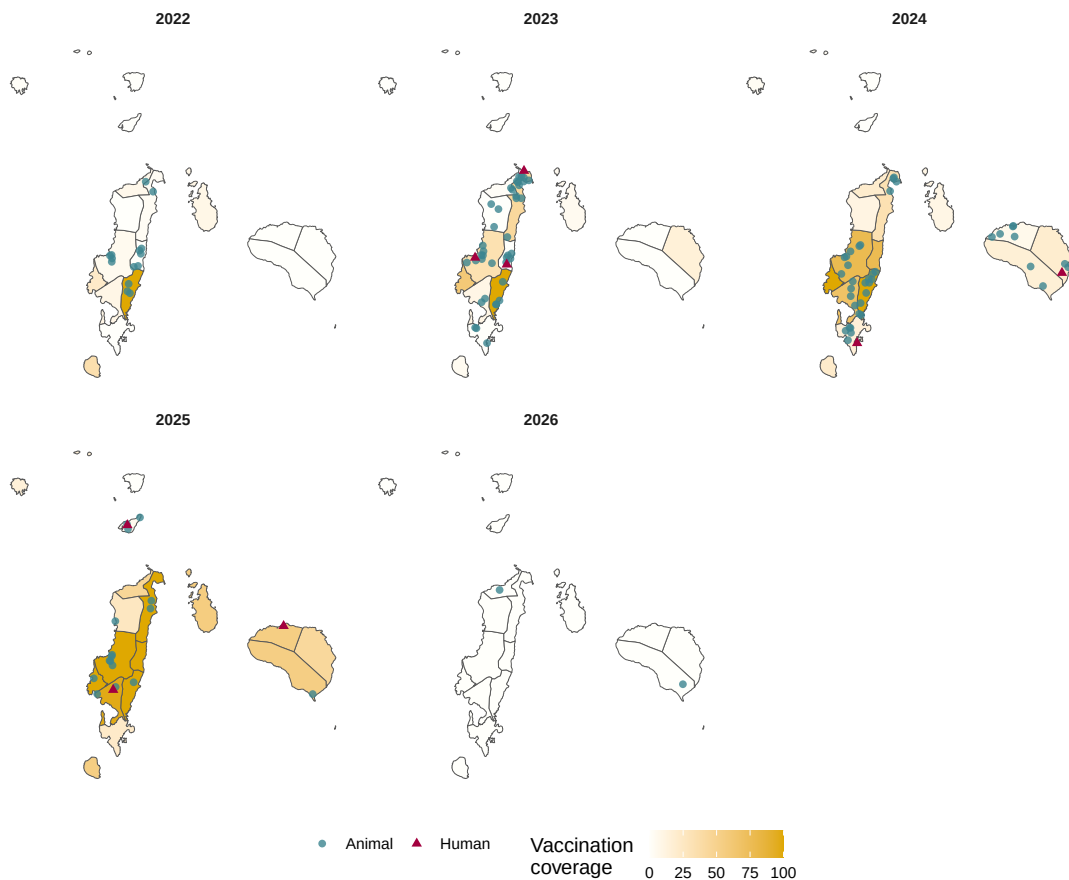


4.2.3 What does the spatiotemporal distribution of these cases suggest about how rabies spread across the province?

Try mapping these cases by year. Use `facet_wrap` again! The islands of interest are Tablas, Sibuyan and Corcuera. You'll need to look at the maps to think about where to put the island

labels! Start by mapping the municipality dataframe to show the vaccination coverage across the municipalities. This is a bit tricky.

💡 Show solution



Note: some vaccination has occurred in 2026, however this data has not yet been released

4.3 Marinduque

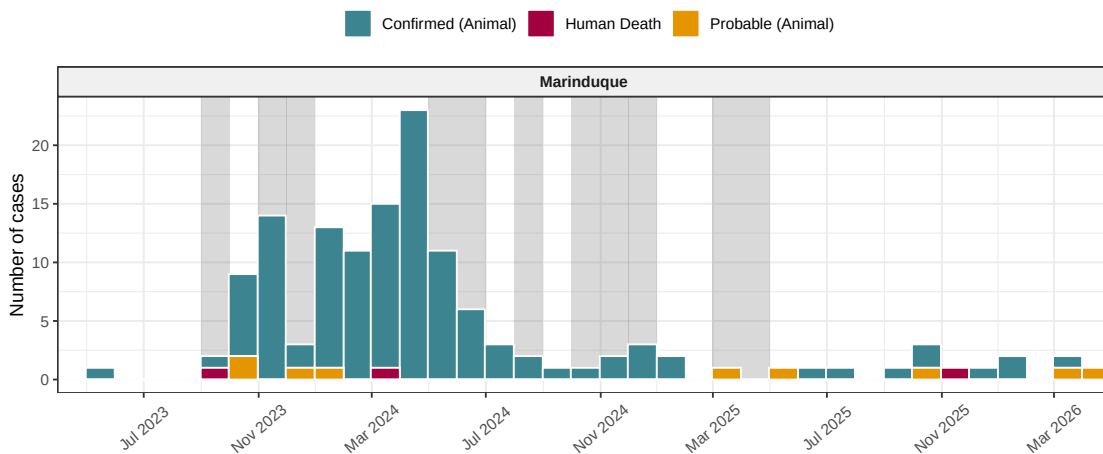
Key questions and things to consider:

- How did the outbreak evolve in Marinduque over time?
- What do these plots tell you about the current surveillance efforts in Marinduque?
- When did the first death occur? And how does this correspond to the increase in dog vaccination efforts?
- How does the outbreak in Marinduque compare to the outbreaks in Oriental mindoro and Romblon provinces?
- How does the human death in Dec 2025 impact your interpretation?

4.3.1 What impact do you think the mass dog vaccination campaigns had on rabies transmission?

You will need to filter the outbreak data to Marinduque Province. Overlay the mass dog vaccination data to the timeseries plot. The mass dog vaccination data will need filtering too!

💡 Show solution



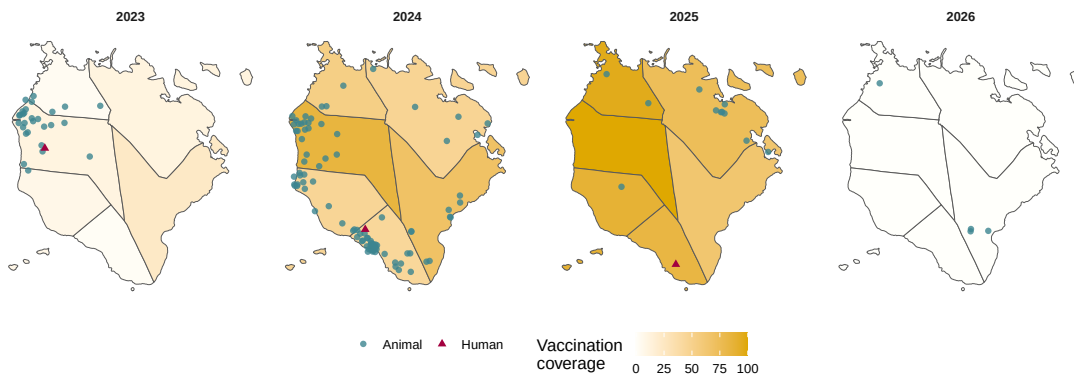
Dates of mass dog vaccination campaigns are shown in grey (a campaign defined as >700 dogs vaccinated in a municipality in a month)

4.3.2 What do you think the prospects are for Marinduque eliminating rabies from the province?

Try mapping these cases by year. Use `facet_wrap` again! Start by mapping the municipality dataframe to show the vaccination coverage across the municipalities. This is a bit tricky.

Elimination requires interruption of rabies transmission in domestic dogs. What do you think about the quality of rabies surveillance on Marinduque? Can you be confident that when you are no longer seeing dog rabies cases, that rabies has been eliminated? or do you think that an absence of detected rabies cases just reflects weak surveillance? Also remember that the incubation period for rabies is on average around one month, but can sometimes be several months long. What does this mean for being confident that rabies has gone?

💡 Show solution



Note: some vaccination has occurred in 2026, however this data has not yet been released

5 Now lets think about what all this means across the MIMAROPA region

💡 Some useful background!

Rabies is spread by the bite of an infected rabid animal and is **invariably** fatal! This is why post-exposure vaccination (PEP) is needed immediately after a rabid dog bite. If a bite victim only attends a hospital after showing signs of rabies it is too late. But only *around one in 5 rabid dog bite victims will develop rabies* if unvaccinated. So for every unlucky rabies patient, four other rabid bite victims will get lucky and survive without developing rabies.

Usually only around one in 3 rabid dogs bite people; a few of these will bite a lot, but many will bite 1 or 2 people. Remember that surveillance only ever provides a partial picture, of the cases observed or confirmed through sample collection, laboratory testing and official reporting. Can you think of reasons why rabid dogs may not be counted? and what about human rabies cases?

Key questions and things to explore:

- Which province do you think has the most robust surveillance? (consider surveillance from the veterinary perspective, the public health perspective and under a One Health approach)
- Do you consider rabies to be endemic in any of the provinces?
- What do you think are the triggers of outbreak responses?
- Why do the ratios of human to animal cases differ in the three provinces?

5.1 How do human deaths compare with animal cases?

A critical question is whether surveillance is sufficient. If we have areas where human deaths occur with relatively few reported animal cases this would suggest under-reporting of animal cases. Therefore, the annual incidence and the ratio of human deaths to animal cases may provide an indication of the surveillance sensitivity and reporting. Try working through the rest of this optional code and see what you learn in the process:

5.1.1 Ratio of human deaths to animal cases

Calculate the ratios of human deaths to animal cases across the provinces? What do these suggest about how the quality of surveillance varies across the different provinces?

💡 Show solution

Province	Animal cases	Human deaths	Deaths per 100 animal cases
Marinduque	129	3	2.3
Oriental Mindoro	76	32	42.1
Romblon	108	8	7.4

5.2 Is rabies an emerging infectious disease threat or a neglected disease of poverty?

Rabies can be viewed as an emerging disease that can cause outbreaks in areas where it was previously absent. But it is also a disease that is entirely vaccine-preventable, that can be eliminated through dog vaccination, whilst human deaths only occur if people at-risk do not access emergency life-saving vaccines after an exposure.

The last decade in MIMAROPA region illustrates all of these different situations. Can you quantify the incidence of human rabies deaths and of rabies in dogs in each province and how this varies by year? What is the highest incidence? and what is the lowest incidence? Taking these extremes, and extrapolating to the entirety of the Philippines, what is the worst case scenario? how many rabies deaths *could* occur across the country if they were all to face the same challenges in rabies control and prevention?

And what if rabies control improved and surveillance was better - what is the best case scenario? What are the obstacles to achieving this best case outcome? and what can be done better that can support the best outcome?!

5.2.1 Population estimates by province

Location	Human population	Dog population
Oriental Mindoro	900,000	200,000
Occidental Mindoro	510,000	110,000
Marinduque	220,000	40,000
Romblon	284,000	55,800

5.2.2 Incidence of human deaths and animal cases

Calculate the annual incidence of human deaths and animal cases by province.

💡 Show solution

Province	Animal cases	Animal incidence (per 1,000 dogs/year)	Human deaths	Human incidence (per 100,000 people/year)
Marinduque	134	0.48	3	0.19
Occidental Mindoro	9	0.01	3	0.08
Oriental Mindoro	79	0.06	38	0.59
Romblon	110	0.28	8	0.40
Overall	332	0.12	52	0.38

5.2.3 How does the annual incidence extrapolate?

Romblon in 2023 and 2025 recorded the highest annual human rabies incidence at 1.06 deaths per 100,000 people. What happens if we extrapolate these numbers to the overall population (population estimate of the Philippines ~113 million)?

Scenario	Rate	Expected_deaths
Regional average	0.38	429
Worst case	1.06	1198

What else can help you to understand what is happening with rabies across MIMAROPA?:

- Sequencing of rabies cases from these provinces reveals important information for understanding rabies spread:
- Cases in Oriental Mindoro are closely related to one another, and distinct from rabies viruses circulating in the other provinces. The cases in Romblon province (Tablas Island) are most closely related to rabies viruses from Region 4A. They cluster into 3 distinct groups suggesting 3 separate introductions from Region 4A. Cases on Sibuyan and Corcuera islands in Romblon province are most closely related to cases from one of the Tablas island clusters. Cases in Marinduque are closely related to one another and appear to have a single ancestor from Region 4A.
- Given this inter-island and inter-regional rabies spread, what recommendations do you have about coordinating control measures and communications for sensitizing communities?

Based on all the available evidence, what would you recommend the municipal, provincial, regional and national One Health teams do next, and why?!